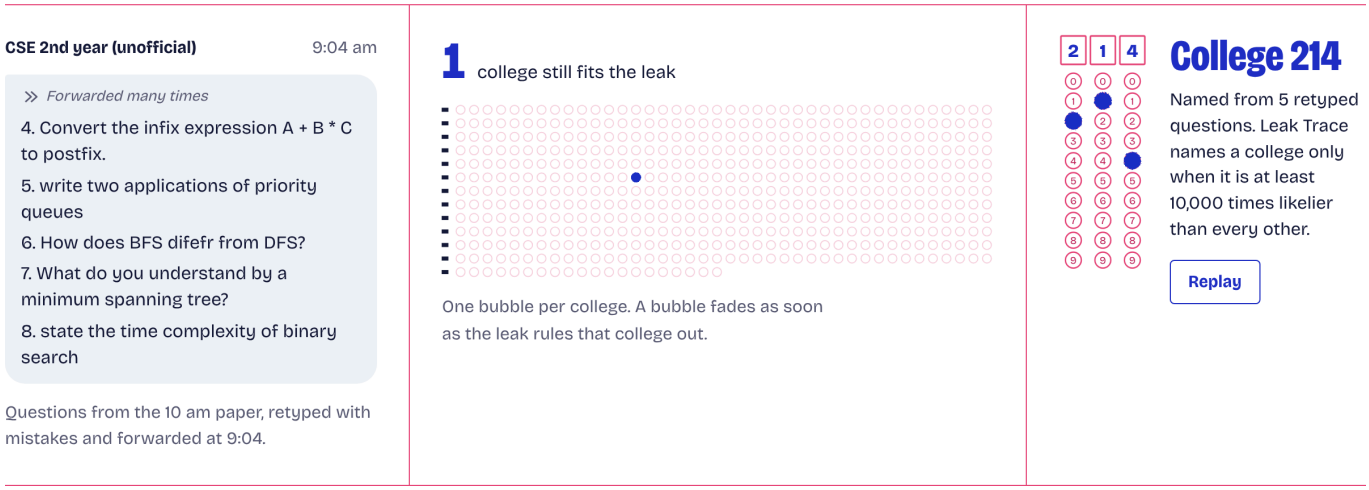
THEME 4 · SHIKSHIT & KAUSHAL BHARAT · TRACK A, PRODUCT & TECHNOLOGY

Which of 500 colleges leaked the paper?

Leak Trace gives every college the same exam with a hidden pattern in it. When a few questions, a photo or even just the answer key turns up on WhatsApp, the pattern points back to the college whose copy leaked. It names a college only at odds of 10,000 to 1, and otherwise gives investigators a ranked shortlist.



From the live demo: five questions retyped with mistakes and forwarded at 9:04. Each line rules out more colleges until one is left, at more than 10,000 to 1.

Problem statement	Stage	Team
Tracing which college leaked a university exam paper	**Working prototype, live demo, research paper**	**Team Leak Trace, Sreyas Institute of Engineering and Technology, Hyderabad**

92%
of leaks of 7 numbered questions named; source in the top 5 from 4 questions

0
innocent colleges named in 135,000 honest leaks; 1 in 148,000 attacks

14
correct answers in a leaked answer key are enough to name the college

1 in 10,000
proven bound, under our model, on a copy from outside the system clearing the naming bar

Try it yourself: be the leaker

The decoder runs in your browser against 500 simulated colleges. Make a leak, retype it, reword it, shuffle it, or leak only the answer key, and see if Leak Trace catches you.

raw.githack.com/SakethSumanBathini/SakethSumanBathini/leak-trace-site/index.html

A leak shows up. Nobody can say where it came from.

Problem statement (59 words)

When a university exam paper leaks on WhatsApp, nobody can tell which of its 100-plus colleges leaked it. Leakers post a few typed or cropped questions without QR codes or printed IDs, so sources surface only by luck or police work, and every student risks a re-exam. It happened at Kannur University and Karnataka State Law University in 2025.

89

suspected paper leaks reported in India in the ten years to 2024, 68 in state exams; at least 48 papers re-tested

4 min

from the PIN reaching a Kannur principal (9:00) to questions on WhatsApp (9:04), April 2025

3 months

for the CBI to attribute the NEET-UG 2026 leak, helped by handwriting examination

1 college

to re-examine, Kannur's sub-committee recommended once the source was known, instead of a re-exam for everyone

How papers reach students. Universities now send each college a password-protected PDF shortly before the exam, and the college prints it (Kannur, Bangalore University, VTU; JNTUH lists electronic distribution). That printing step is where several leaks below began.

When	Where	What happened	How the source was found
Jul 2026	**Uttarakhand Technical Univ.**	B.Tech questions on WhatsApp matched the paper 90–95%; two exams cancelled.	**An anonymous email;** a professor at an affiliated college arrested.
May 2026	**NEET-UG (national)**	Cancelled after a "guess paper" matched 45 of 108 chemistry questions.	**Months of forensics;** three NTA subject experts (CBI).
Jan 2026	**PJTSAU, Telangana**	Papers for six exams reached in-service students.	**By chance.** The VC noticed unusually high marks; CCTV showed a student finishing in ten minutes.
Jan 2026	**Karnataka SSLC prep**	Downloaded with teachers' logins, sold for ₹200–500.	**Police investigation;** six teachers arrested.
Apr 2025	**Kannur University**	PIN at 9:00; 4–5 questions on WhatsApp at 9:04; exam at 10:00.	**By luck:** an exam squad caught a student copying.
Jan 2025	**KSLU, Karnataka**	Sold as "two, five, or 10 questions".	**Phone forensics,** back to a vice-principal.

Who is affected

- **Students** at every college that wrote the paper face a re-exam, delayed results and suspicion.
- **Honest colleges and staff** are suspects until the source is found.
- **Universities** pay to re-set, reprint, re-invigilate and re-value, and lose credibility.
- **Investigators** start with no lead and trace phones for days or weeks.

Why it matters now

- The 2026 amendment to the Public Examinations Act toughened penalties and set 60-day investigations; investigators need fast, specific leads.
- NEET moves to computer-based testing from 2027, and the exam-reform task force is looking at last-minute paper generation. Neither says which centre leaked.

Sources on page 10

Every current fix breaks when someone retypes five questions.

Fix	Used by	Where it breaks
Printed QR code or serial number	Haryana board, Telangana SSC, NIOS	Gone the moment questions are retyped
Hidden code readable from a photo	West Bengal board	Gone the moment questions are retyped
Centre-code watermark	CBSE (2018), vendors	Gone the moment questions are retyped
A few reordered sets	CBSE, NEET booklet codes	4–8 sets cannot single out one college among hundreds
OTP login and access logs	Paper-delivery portals (2,000+ colleges)	Shows who opened the file, not who leaked it
Leak Trace	**Our entry**	**Survives retyping and rewording, tells 500 colleges apart, states its error rate**

What is new

- **The identity is in the content, not on the page.** Question order, approved wordings, option order, numbers and answer letters survive retyping, cropping and photos.
- **Built for hundreds of colleges.** Codes with proven distances keep every pair of colleges apart, so a handful of questions is enough.
- **Marks a leaker cannot cheaply remove.** The values in numerical questions and the letter of each right option vary by college; changing them makes the leaked answers wrong.
- **States its own error rate, and refuses to guess.** Names a college only at 10,000 to 1. Proven under our model: a copy from outside the system clears the naming bar at most 1 time in 10,000 (2 in 10,000 with the two-copy check).
- **Protects itself.** A key split among three officials, no stored table and a signed fingerprint mean no single official can frame a college or swap a paper undetected.

What is not new, and we say so

Giving each recipient a slightly different copy is the old canary trap. An expired IBM patent (filed 2003, granted 2007) used variant questions as a "version identifying code", and vendors and boards print centre codes. None of these traces a retyped, partial leak or an answer key, or states how often it would blame the wrong college.

> **Our search** (30 Oct 2026) of research papers, patents, products and news found no system that traces a leaked paper's source from retyped text or an answer key with a stated error rate. The closest proposals hide a logo in the image (2019) or match a leaked paper exactly against signed sets (2021).

Built on solid ground

Fingerprinting codes (Boneh–Shaw 1998; Tardos 2008), Reed–Solomon codes, sequential likelihood ratios (Wald 1945; Ville 1939), Shamir secret sharing, Merkle trees and Ed25519 signatures; and the psychometrics of order effects, answer positions and item variants.

Same questions, same marks. A different pattern for every college.

Only one step of today's delivery changes: the moment the university makes each college's PDF.

BEFORE THE EXAM

1

Setter writes one paper

and approves two or three wordings per short question, value sets for numerical questions, and which questions may move.

2

Two of three officials unlock

The secret key is split three ways; any two derive the 500 codewords. No table is stored.

3

500 PDFs, keys and a seal

Each college's version, its answer key, and a fingerprint of all 500 PDFs signed by two officials and sent to every principal.

4

Delivery as today

Each college downloads with its PIN or OTP and prints its own copy. Nothing changes for students.

AFTER A LEAK

5

Leak spotted

A few retyped questions, a photo or crop, or an answer key on WhatsApp.

6

Read it

OCR reads photos; fuzzy matching ties each line, typos and all, to its master question and wording.

7

Score all 500

A likelihood for every college, every pair of colleges (mixed copies) and a copy from outside.

8

Lead, with odds

Names a college at 10,000 to 1 with the features that matched; otherwise a ranked shortlist.

Q.P. code: SET 214

B.Tech Degree Examination, November 2026

Third Semester: Data Structures

Time: 3 hours — Max. marks: 70

PART A

Answer all questions. Each question carries 2 marks.

1. State what the height of a binary tree is.
2. Define a circular linked list.
3. Define an abstract data type and give one example.
4. Convert the infix expression A + B * C to postfix.
5. Write two applications of priority queues.
6. How does BFS differ from DFS?
7. What do you understand by a minimum spanning tree?
8. State the time complexity of binary search.
9. Compare a stack and a queue.
10. Define hashing and name one collision resolution technique.

Printed for College 214

Q.P. code: SET 037

B.Tech Degree Examination, November 2026

Third Semester: Data Structures

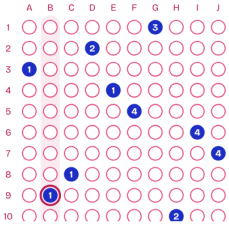
Time: 3 hours — Max. marks: 70

PART A

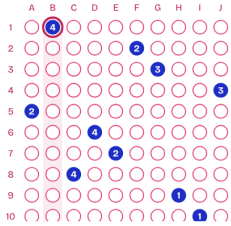
Answer all questions. Each question carries 2 marks.

1. How does a stack differ from a queue?
2. Give two applications of a priority queue.
3. State what the height of a binary tree is.
4. Explain the term minimum spanning tree.
5. What is an abstract data type? Give one example.
6. What do you mean by a circular linked list?
7. Convert A + B * C from infix to postfix form.
8. Write down the time complexity of a binary search.
9. What is hashing? Name one collision resolution technique.
10. Differentiate between BFS and DFS.

Printed for College 037



College 214's pattern. No other college has this one.

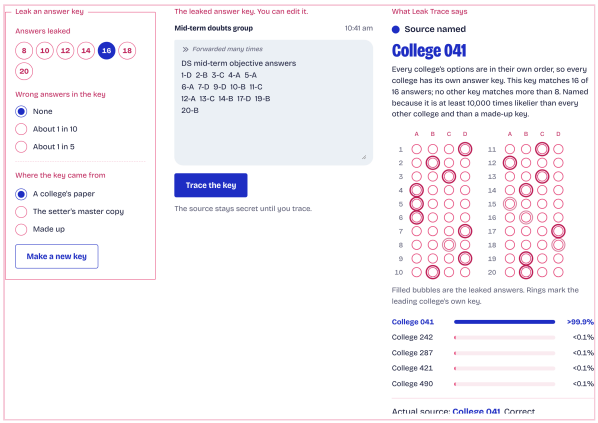


College 037's pattern. No other college has this one.

Part A as printed for two colleges: the same ten questions and marks, in a different order and with different approved wordings (question B highlighted). The grids read like an OMR sheet: columns are questions, rows are printed positions, the digit is the wording. No other college has either pattern: any two colleges share at most 4 of the 10 positions.

From easy to erase, to costly to erase.

What the leaker does	Order questions, labels, options	Wording approved variants	Numbers and answers values, right-option letters
Retypes it, with mistakes	● Survives	● Survives	● Survives
Sends a photo or a crop	● Survives	● Survives	● Survives
Drops or changes the numbers	● Order of lines counts	● Survives	● Survives
Shuffles the lines	○ Erased	● Survives	● Survives
Rewords every question, even with AI	● Survives	○ Erased	● Survives
Rewords and shuffles	○ Erased	○ Erased	● Survives



The answer-key demo: a 16-answer key leaked on WhatsApp is matched against 500 colleges' keys and the setter's master key. Filled bubbles are the leak; rings are the named college's key.

Even an answer key gives the college away

For MCQ papers whose key may leak, each college's options are arranged so the right answers sit under its own letters. The keys come from a quaternary code: **any two colleges' keys differ in at least 12 of 20 answers**, and each differs from the setter's master key in at least 12, so a leaked master key points upstream. Keys are balanced (each letter 3–7 times, 8–12 right answers in the middle positions), so no college's key is lopsided. A key leaked from one college is also mostly wrong everywhere else.

A proven bound on false names

Under our model of copies made outside the system, such a copy clears the bar for naming a college at most 1 time in 10,000 (2 in 10,000 with the two-copy check), however often investigators re-run it as more lines arrive (Markov's and Ville's inequalities). Wrong names for real leaks are measured, not bounded: none in our tests.

$$\mathbb{P}(\text{ some college reaches } 10{,}000 \times 500\ ) \le 1 / 10{,}000$$

Numbers work the same way

Faculty approve value sets chosen to be of equal difficulty, to be calibrated in a pilot; keys are generated per set. Reworded or not, a leaked question keeps its numbers, or its answer changes.

SET 041

A simply supported beam of span **6 m** carries a point load of **12 kN** at mid-span. Find the maximum bending moment.

SET 213

A simply supported beam of span **5 m** carries a point load of **16 kN** at mid-span. Find the maximum bending moment.

With 16 approved sets per question, six reworded and shuffled numerical questions name the source in 83% of simulated leaks; with 64 sets, four questions in 89%. No innocent college named in 63,000 such leaks.

Seven numbered questions name the source 92% of the time.

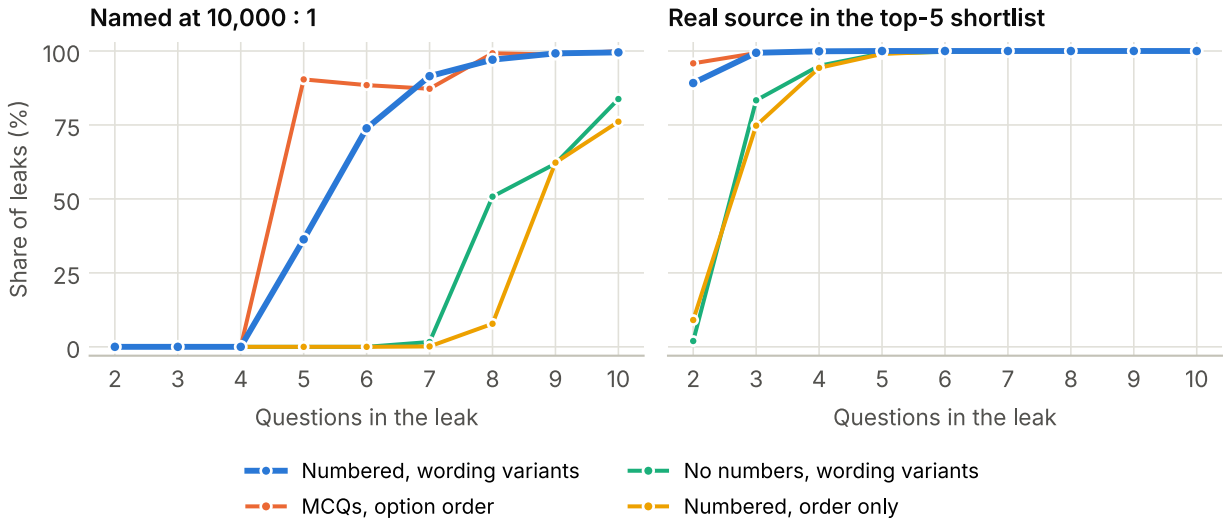
500 simulated colleges, a Data Structures paper, typing damage and reading noise; a college is named only at 10,000 to 1. **All results so far are synthetic**; real leaks made by real people are the next test.

0
innocent colleges named in 135,000 honest single-college leaks

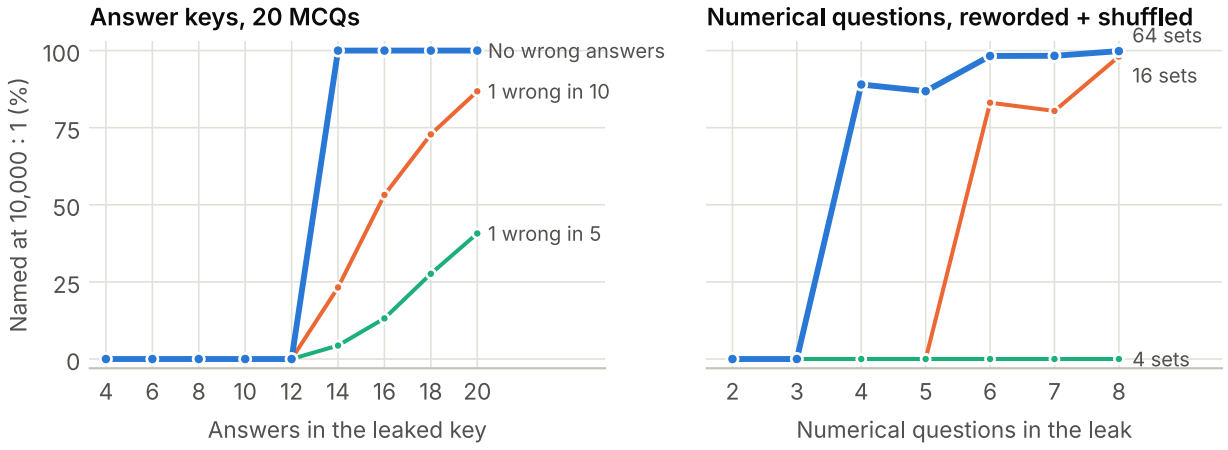
1
innocent college named in a 148,000-leak attack test (an outside copy that matched by chance)

30 of 30
synthetic phone photos showing 5–6 questions traced, through OCR, to the right college

0
innocent colleges named in 171,000 answer-key and numerical-question leaks



Whole-paper simulation, 3,000 leaks per point. Left: named at 10,000 to 1. Right: the real source in a top-5 shortlist (ties counted against us). From 4 numbered questions with wording variants, the source is in the top 5 in 99.9% of leaks.



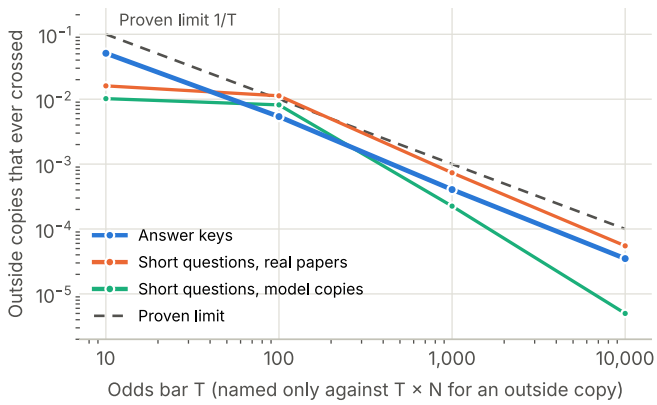
Layer 3. Left: a leaked key with 14 correct answers is named every time; with one wrong answer in ten, 87% of 20-answer keys are named. Right: numerical questions reworded and shuffled so only the values survive: with 16 approved value sets, 6 questions name the source in 83% of leaks and 8 in 98%.

148,000 attacks. One wrong name.

We attacked the decoder that runs in the live demo, 2,000 leaks per setting. Tricks can stop it naming anyone; they did not make it blame an innocent college, except once, for a copy made outside the system that matched a college's pattern by chance. Before we hardened the decoder, the same leaks named eight.

What the leaker did	Named from	Source in top 5 (6 lines)	Innocent named
Kept the printed numbers	5 lines	100%	**0 of 12,000**
Left the numbers out	7 lines	100%	**0 of 12,000**
Renumbered the lines 1, 2, 3	7 lines	100%	**0 of 12,000**
Reworded every question, kept numbers	holds back (61% at 10)	99.8%	**0 of 12,000**
Reworded every question, no numbers	holds back	56.3%	**0 of 12,000**
Shuffled the lines, no numbers	holds back	31.4%	**0 of 12,000**
Mixed two colleges' copies, numbered	10 lines	>99.9%	**0 of 6,000**
Sent a copy from outside the 500	–	–	**1 of 32,000**
All 74 settings			**1 of 148,000**

"Named from": smallest leak named in at least 95% of cases. Rewording and shuffling are what layer 3 is for. Under heavy typing damage (15% of characters), 5 numbered lines are still named 99.2% of the time.



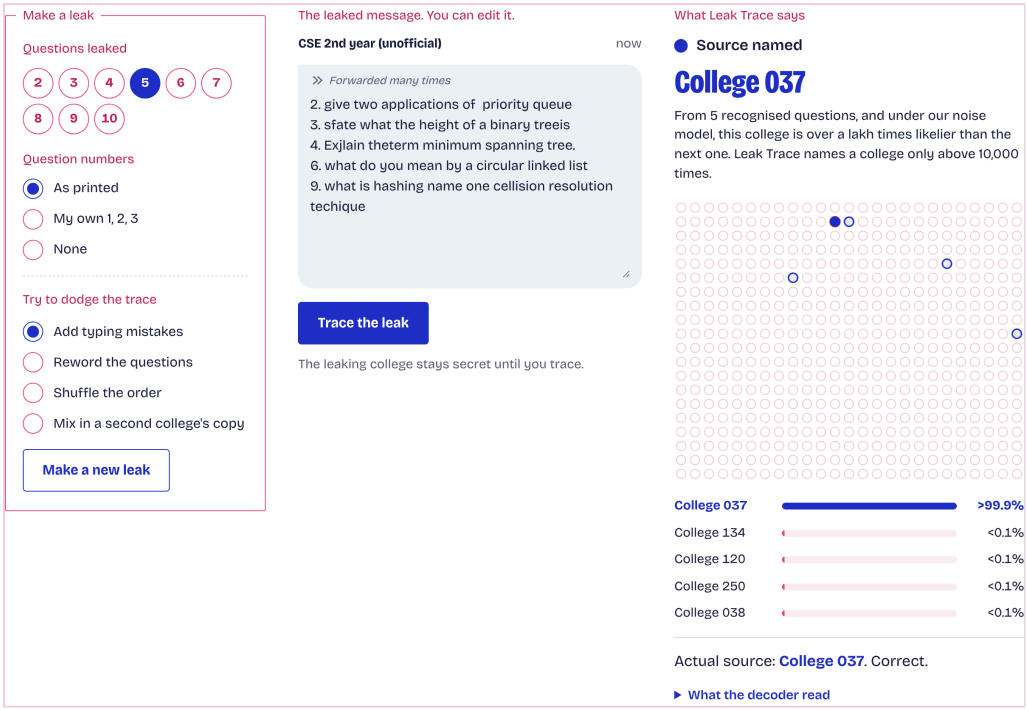
The proof, checked on 800,000 outside copies

We read 400,000 outside short-question copies (half following the model, half real permutations) and 400,000 outside answer keys one line at a time. At the 10,000-to-1 bar, 11 of the 200,000 realistic copies and 14 of the answer keys ever crossed it, inside the bound. Crossing it is not enough to be named: the leader must also beat every other college by 10,000 to 1. In the attack test, the full decoder named 1 of 32,000 outside copies.

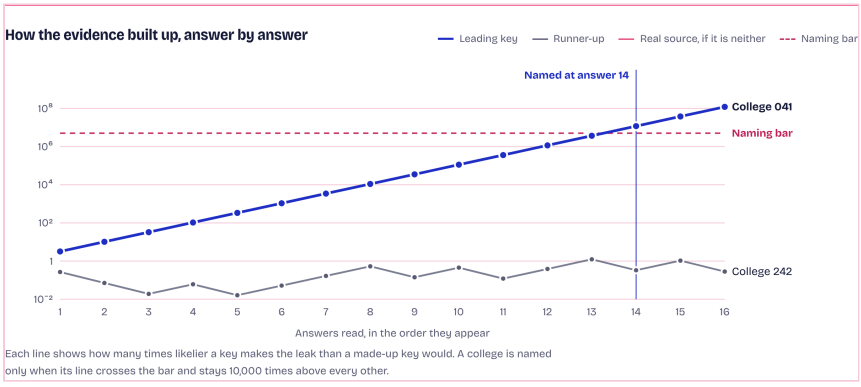
Honest detail: at a 100-to-1 bar, real papers crossed slightly more often than the proof's model allows (1.13% against 1%), because a real paper's numbers never repeat. At the bar we use, they stay inside the bound.

Be the leaker. See if Leak Trace catches you.

The live demo runs the real decoder in the browser against 500 simulated colleges. Choose how many questions to leak, drop or change the numbers, add typing mistakes, reword, shuffle or mix two colleges' copies; or leak only the answer key. Then trace it, and run 200 random leaks at once.



The demo's main bench: the leak maker, the leaked WhatsApp message (editable), and the verdict with the 500-college grid and the top-5 shortlist.



The evidence meter: odds for the leading college and the runner-up after each line or answer, against the naming bar.

What is built

Codebooks and codeword derivation from a split key, page rendering for tests, OCR on photos, fuzzy matching, the decoder with abstention and mixed-copy detection, answer-key tracing, key custody (12 of 12 checks) and the live demo.

Technology

Python, NumPy, Tesseract OCR, OpenCV, RapidFuzz, cryptography (Shamir, HMAC-SHA256, Ed25519); JavaScript for the in-browser decoder. All open source. Decoding takes under a second.

Not done yet

Real leaks from real people (leak lab, November), a pilot in one college's internal exams, and Indian-language papers.

A re-exam at one college instead of all.

Who benefits, and how

- **Students:** when the source is known, only that college re-sits; Kannur's sub-committee recommended exactly that. Thousands of honest students keep their results.
- **Honest colleges and staff:** cleared quickly instead of sitting under suspicion for weeks.
- **Universities:** a targeted re-exam costs a fraction of a university-wide one, and results are not delayed for everyone.
- **Investigators:** a specific lead, with odds, within minutes; useful under the 60-day investigation limit.
- **Prevention:** an insider who knows a leak points home has a strong reason not to leak. Deterrence may be the largest effect, at no extra cost.

Who could adopt it

Controllers of Examinations at affiliating universities (Osmania, JNTUH, Kakatiya, Kannur, KSLU, Bangalore University, VTU, Anna University); autonomous colleges for internal exams; state school and pre-university boards; question-paper delivery vendors; TSCHE, UGC and AICTE as a standard; the National Testing Agency for computer-based tests; police cyber crime units.

Estimated cost

Software licences (all open source)	**₹0**
Build to pilot-ready: 4 students, 3 months, if stipends of ₹10,000 a month were paid	**≈ ₹1.2 lakh**
Hosting: one small cloud server, or nothing extra on an existing exam server	**₹2,000–2,500 / month**
Per exam cycle: no extra printing; setter time for 2–3 wordings or value sets per question	**1–2 hours per paper**
One-semester pilot in one college	**< ₹25,000**

Who pays: the university's exam budget, or a delivery vendor adding it as a feature. Assumptions are ours; no figure is from an official source. Against this: a university-wide re-exam means setting, printing, invigilating and valuing again at every college.

Rollout

Nov 2026	Campus leak lab with classmates; student survey
Dec–Jan	Per-college PDF generator, answer keys, faculty guide for variants
Next semester	Pilot in one college's internal exams, one version per hall; marks published per version
2027	One affiliating university exam series; Telugu and other Indian-language papers

Scale

One adoption covers all of a university's colleges; more colleges need about one more leaked line, because the bar rises with their number. Delivery vendors already reach 2,000+ colleges, and for computer-based tests each centre can get its own order at no cost.

Different order, same exam. A lead, not a verdict.

Fairness

- Reorder only where we expect order not to change difficulty; keep related questions together and easy-to-hard order fixed.
- Faculty approve every wording and value set, to be calibrated in a pilot; answer keys generated per version.
- Balanced answer keys for every college.
- Mean marks per version published after each exam.

Safeguards

- Master key split among three officials; any two needed.
- No stored college-to-pattern table.
- A signed fingerprint of all 500 PDFs reaches every principal before the exam, so no paper can be swapped later.
- Every result carries its odds; the university's inquiry decides.

Limits, stated plainly

- All evaluation is synthetic so far.
- Leaks from setters or the press come before versions exist. A leaked answer key that matches the master says "look upstream"; for question text Leak Trace holds back.
- A leaker who rewords and shuffles every short question leaves only numbers and answers to trace; otherwise Leak Trace holds back.
- English only for now.

Support we are asking for

- **Pilot access:** one internal exam cycle in our college, and a sandbox of one university's per-college PDF delivery.
- **Mentors:** an exam-branch official; a coding-theory or cryptography expert; a legal mentor on using an attribution in an inquiry.
- **Data:** anonymised past leaks and real WhatsApp texts and screenshots.
- **Technical:** cloud credits, Indian-language OCR data, a security review; IP guidance; under ₹25,000 for a pilot.

Read more

Research paper (15 pages): the method, the codes, the proof and every experiment, including where it fails.
`…/leak-trace-site/leak-trace-paper.pdf`
Code and data: `github.com/SakethSumanBathini/SakethSumanBathini`

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